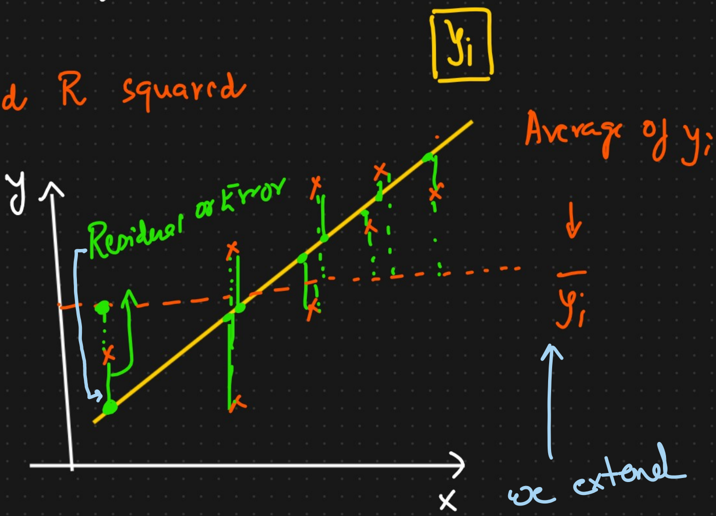
Performance Metrics Used In Linear Regression

① R squared ② Adjusted R squared



$$R\text{squared} = 1 - \frac{SS_{Res}}{SS_{TOTAL}}$$

$\hat{y}_i$

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

we extend this line on the basis of our average predicted number.

$$= 1 - \frac{\text{Small number}}{\text{Bigger number}}$$

see the difference btw our predicted point & $\bar{y}_i$ line [to high Compare to high Compare to Actual point & $\bar{y}_i$ difference]

= 1 − Smaller number → 1

$\approx$ 1

0.70 ⇒ 70%

0.85 ⇒ 85%

0.90 ⇒ 90%

The Value more towards the 1; Model is more Accurate.

⇓

Accurate My Model Is?

{Overfitting, Underfitting}

R squared ↑↑↑

Size of the house ↑ Price ↑

+ve Correlation

② Adjusted R squared

Dataset → (Price)

No. of bedroom ↑ Price ↑

Gender	Size of the house	No. of bedrooms	Location	Price

+ve correlation

Assume

R squared = 75% ⇒ 0.75

This is the problem of R Squared

R squared ⇒ 80% ⇒ 0.80

R squared ⇒ 85% ⇒ 0.85

R squared ⇒ 87% ⇒ 0.87

I. Look at the all features:
Except the Gender feature other are related to price feature. So still if the feature is not co-related to o/p feature is still directly add to the R-squared value, but with small number increase

Instead of R-squared use ↓

$$\text{Adjusted R Squared} = 1 - \frac{(1-R^2)(N-1)}{N-P-1}$$

$p=2 \quad R^2 = 90\% \quad R^2 \text{ adjusted} = 86\%$

$p=3 \quad R^2 = 92\% \quad R^2 \text{ adjusted} = 82\%$

Assumeable value

If we derive for that R^2 adj.

N = No. of data points

P = No. of Independent features